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## Sensing Device

### Abstract


A sensing device is disclosed. The sensing device according to an embodiment of the present disclosure includes an infrared sensor; a condenser lens located on a front side of the infrared sensor; and a cover located on a front side of the condenser lens to cover the condenser lens, and having a hole that is smaller than the condenser lens.



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## Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a sensing device. More particularly, the present disclosure relates to a sensing device installed in a restricted area such as dangerous goods storage, guardrails and the like to detect an intruder and provide an alarm.

[0002] The present application claims priority to Korean Patent Application No. 10-2023-0023952 filed on Feb. 22, 2023 in the Republic of Korea, the disclosure of which is incorporated herein by reference.

BACKGROUND ART

[0003] There are some restricted facilities or areas from which humans are not allowed to enter or facilities in which entry is not allowed for a certain period of time. For example, unauthorized persons are not allowed to enter facilities such as dangerous goods storage and guardrails or construction sites such as scaffolding sites to prevent safety incidents.

[0004] One of methods for banning entry to prohibited or restricted areas is to place signs indicating prohibited areas such as traffic cones in the corresponding facilities or areas to provide visual indication that passing or entering is prohibited.

[0005] However, in frequent instances, those who did not recognize the indication may enter the restricted areas and be exposed to the risk of safety incidents.

[0006] To solve this problem, systems for preventing safety incidents or illegal behaviors have been proposed, in which a warning sound is issued when a human is detected in a restricted area using an infrared sensor capable of detecting humans.

[0007] A typical example of the infrared sensor for detecting humans is a Passive Infrared (PIR) sensor. The PIR sensor may detect infrared light emitted from human body. Additionally, in general, the detection angle of the PIR sensor is about 120°. However, when the PIR sensor having the detection angle of 120° is used, a very wide detection range may be set. As a consequence, the detection system may output a warning for a behavior or action that is not actually an entry to the restricted area or break-in.

SUMMARY

Technical Problem

[0008] The present disclosure is directed to providing a sensing device including a structure for effectively limiting a detection range.

[0009] The present disclosure is further directed to providing a sensing device including a structure for maintaining sensitivity over the reduced detection range.

[0010] The present disclosure is further directed to providing a sensing device having a structure for enabling cover replacement to variously adjust the detection range.

[0011] The present disclosure is further directed to providing a sensing device that is easy to install.

Technical Solution

[0012] To achieve the above-described objective, a sensing device according to an embodiment of the present disclosure includes an infrared sensor; a condenser lens located on a front side of the infrared sensor; and a cover located on a front side of the condenser lens to cover the condenser lens, and having a hole that is smaller than the condenser lens.

[0013] Additionally, the cover may include a portion having an increasing inner diameter toward a rear side.

[0014] Additionally, a distance between the hole and the infrared sensor may be larger than a diameter of the hole.

[0015] Additionally, the cover may extend rearwards.

[0016] Additionally, an inner surface of the cover may have an emissive structure.

[0017] Additionally, the condenser lens may have a frontward protruding shape, and the cover may accommodate at least a portion of the condenser lens.
[0018] Additionally, the cover may include a shield wall protruding from an inner surface and covering the condenser lens.
[0019] Additionally, the shield wall may include a shield hole having a smaller diameter than the hole.
[0020] Additionally, the cover may be rotatable in a direction in which the infrared sensor is oriented.
[0021] Additionally, the sensing device may further include a body at which the infrared sensor is installed; and a warning unit disposed in the body and configured to output sound or light in response to infrared light detected through the infrared sensor.
[0022] Additionally, the sensing device may further include a body at which the infrared sensor is installed; and a pair of sights protruding from an upper surface of the body.
[0023] Additionally, the sensing device may further include a body at which the infrared sensor is installed; and a coupling bar extending below the body.
[0024] Additionally, the sensing device may further include a mount that is coupled to the coupling bar.
Advantageous Effects
[0025] According to at least one of the embodiments of the present disclosure, it may be possible to provide the sensing device including the structure for effectively limiting the detection range to improve accuracy.
[0026] According to at least one of the embodiments of the present disclosure, it may be possible to provide the sensing device including the structure for maintaining sensitivity over the reduced detection range.
[0027] According to at least one of the embodiments of the present disclosure, it may be possible to provide the sensing device having the structure for enabling cover replacement to variously adjust the detection range.
[0028] According to at least one of the embodiments of the present disclosure, it may be possible to provide the sensing device that is easy to install.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0029] The accompanying drawings illustrate exemplary embodiments of the present disclosure and together with the following detailed description, serve to provide a further understanding of the technical aspect of the present disclosure, and thus the present disclosure should not be construed as being limited to the drawings.
[0030] FIG. **1** is a perspective view of a sensing device according to an embodiment of the present disclosure.
[0031] FIG. **2** is a partial perspective view of a sensing device according to an embodiment of the present disclosure.
[0032] FIG. **3** is a partial exploded view of a sensing device according to an embodiment of the present disclosure.
[0033] FIG. **4** is a cross-sectional perspective view of FIG. **2**, taken along the line A-A′.
[0034] FIG. **5** is a cross-sectional view of FIG. **2**, taken along the line A-A′.
[0035] FIG. **6** is a diagram showing a detection area of a sensing device according to an embodiment of the present disclosure.
[0036] FIG. **7** is a diagram showing an inner surface of a cover according to an embodiment of the present disclosure.

[0037] FIG. **8** is a partial perspective view of a sensing device according to an embodiment of the present disclosure.
[0038] FIG. **9** is a cross-sectional perspective view of FIG. **8**, taken along the line B-B′.
[0039] FIG. **10** is a cross-sectional view of FIG. **8**, taken along the line B-B′.
[0040] FIG. **11** is a partial diagram of a sensing device according to an embodiment of the present disclosure.
[0041] FIG. **12** is a diagram showing a detection range of a sensing device according to an embodiment of the present disclosure.
[0042] FIG. **13** is a diagram showing a sensing device according to an embodiment of the present disclosure.
[0043] FIG. **14** is an exploded view of a configuration for installing a sensing device according to an embodiment of the present disclosure.
[0044] FIG. **15** is a diagram showing an installation example of a sensing device according to an embodiment of the present disclosure.
[0045] FIG. **16** is a diagram showing a plurality of sensing devices installed according to an embodiment of the present disclosure.
[0046] FIG. **17** is a diagram showing a sensing device according to an embodiment of the present disclosure.
[0047] FIG. **18** is an exploded view of a configuration for installing a sensing device according to an embodiment of the present disclosure.
[0048] FIG. **19** is a diagram showing an installation example of a sensing device according to an embodiment of the present disclosure.
[0049] FIG. **20** is an exploded view of a configuration for installing a sensing device according to an embodiment of the present disclosure.
[0050] FIG. **21** is a diagram showing an installation example of a sensing device according to an embodiment of the present disclosure.

## DETAILED DESCRIPTION

[0051] Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Prior to the description, it should be understood that the terms or words used in the specification and the appended claims should not be construed as being limited to general and dictionary meanings, but rather interpreted based on the meanings and concepts corresponding to the technical aspect of the present disclosure on the basis of the principle that the inventor is allowed to define the terms appropriately for the best explanation.
[0052] Therefore, the embodiments described herein and the illustrations shown in the drawings are exemplary embodiments of the present disclosure to describe the technical aspect of the present disclosure and are not intended to be limiting, so it should be understood that a variety of other equivalents and modifications could have been made thereto at the time that the application was filed.
[0053] FIG. **1** is a perspective view of a sensing device according to an embodiment of the present disclosure. FIG. **2** is a partial perspective view of the sensing device according to an embodiment of the present disclosure. FIG. **3** is a partial exploded view of the sensing device according to an embodiment of the present disclosure. FIG. **4** is a cross-sectional perspective view of FIG. **2**, taken along the line A-A′. FIG. **5** is a cross-sectional view of FIG. **2**, taken along the line A-A′.
[0054] Referring to FIGS. **1** to **5**, the sensing device according to an embodiment of the present disclosure may include an infrared sensor **100**, a condenser lens **200** and a cover **300**. The infrared sensor **100** may be configured to detect infrared light incident from the external environment. For example, the infrared sensor **100** may be a Passive Infrared (PIR) Sensor.
[0055] The infrared sensor **100** may detect infrared light emitted from human body. In general, the infrared sensor **100** may have a detection angle of about 120°. The infrared sensor **100** may have a detection range oriented in the frontward direction or +X axis direction. The infrared sensor **100**

may be installed in a body **400**.

[0056] The condenser lens **200** may be located on the front side of the infrared sensor **100** or in +X axis direction. The condenser lens **200** may converge infrared light incident from the external environment to a focal point. For example, the condenser lens **200** may be a Fresnel lens. In this instance, the infrared sensor **100** may be located at the focal point of the condenser lens **200**. The condenser lens **200** may be coupled, fastened, fixed or assembled to the infrared sensor **100**.

[0057] The cover **300** may be disposed on the front side of the condenser lens **200** or in +X axis direction. The cover **300** may cover at least a portion of the condenser lens **200**. The cover **300** may be made of a material having the ability to shield light including infrared light. For example, the cover **300** may be made of a polymer.

[0058] Additionally, the cover **300** may have a hole **301**. In particular, the cover **300** may have a hollow extending in the front-rear direction (X axis direction). Additionally, the hollow may be open at the front end and the rear end. Here, the hole **301** may refer to the front opening of the hollow.

[0059] Additionally, the hole **301** may face the condenser lens **200**. For example, the hole **301** may be circular in shape. In this instance, the cover **300**, the condenser lens **200** and the infrared sensor **100** may be aligned such that the center of the hole **301**, the center of the condenser lens **200** and the center of the infrared sensor **100** are located on a straight line.

[0060] The hole **301** may be smaller than the condenser lens **200**. Alternatively, a diameter L**2** of the hole **301** may be smaller than a diameter L**3** of the condenser lens **200**. In particular, the condenser lens **200** may be inserted into the opening at the rear end of the cover **300**. In this instance, the hole **301** at the front end of the cover **300** may be smaller than the condenser lens **200** inserted into the rear end of the cover **300**.

[0061] The cover **300** may be configured to shield at least some of infrared light entering the condenser lens **200**. Additionally, as the diameter L**2** of the hole **301** is smaller, the detection angle or the detection range may be smaller.

[0062] For example, the detection angle of the infrared sensor **100** may reduce from 120° down to 10° due to the cover **300**. Additionally, the hole **301** may have various shapes including a polygonal shape such as triangular, square, rectangular shapes as well as semicircular, oval and elliptical shapes. The cover **300** may be coupled, fastened, fixed or assembled to the condenser lens **200**.

[0063] According to the above-described configuration, the cover **300** may limit the detection range of the infrared sensor **100**. Thus, the detection range of the infrared sensor **100** may reduce. However, even though the detection range of the infrared sensor **100** reduces, due to the condenser lens **200**, sensing sensitivity in the detection range may be maintained or hardly reduced. The user may precisely set the detection area using the sensing device of the present disclosure.

[0064] According to the above-described configuration of the present disclosure, it may be possible to effectively reduce the detection angle without using the thick condenser lens **200** as the component. For example, it may be necessary to use the very thick condenser lens **200** to reduce the detection angle of the infrared sensor **100** down to about 10° without the cover **300**. However, when the very thick condenser lens **200** is used, sensing sensitivity or sensing accuracy may be low.

[0065] Referring to FIGS. **1** to **5**, the cover **300** of the sensing device according to an embodiment of the present disclosure may extend rearwards or in-X axis direction. The cover **300** may extend to the condenser lens **200** or the infrared sensor **100**.

[0066] According to the above-described configuration of the present disclosure, the infrared sensor **100** may detect only infrared light passing through the hole **301**. The cover **300** may shield infrared light entering the condenser lens **200** or the infrared sensor **100** through other path than a path passing through the hole **301**. Accordingly, it may be possible to improve accuracy of the infrared sensor **100**.

[0067] Referring to FIGS. **1** to **5**, the cover **300** of the sensing device according to an embodiment of the present disclosure may include a portion having an increasing inner diameter L**4** as it goes in the rearward direction or −X axis direction. Alternatively, a cross section of the cover **300** perpendicular to X axis direction may include a portion having an increasing diameter L**4** as it goes in the rearward direction or −X axis direction. Additionally, at least a portion of the cover **300** may include a region having no increasing inner diameter as it goes in the rearward direction or −X axis direction. When the hole **301** is circular in shape, the cover **300** may have a conic shape. Additionally, when the hole **301** is polygonal in shape, the cover **300** may have a polygonal prism shape.

[0068] According to the above-described configuration of the present disclosure, infrared light passing through the hole **301** may stably enter the condenser lens **200**. Accordingly, it may be possible to keep sensitivity of the infrared sensor **100** high.

[0069] FIG. **6** is a diagram showing the detection area of the sensing device according to an embodiment of the present disclosure. Referring to FIGS. **5** and **6**, a distance L**1** between the hole **301** and the infrared sensor **100** of the sensing device according to an embodiment of the present disclosure may be larger than the diameter L**2** of the hole **301**. In this instance, the target detection range Z**1** of the sensing device may be set to a distance D**1** or less and a width D**2** or less in the frontward direction or +X axis direction from the hole **301** of the cover **300**. The distance L**1** between the hole **301** and the infrared sensor **100** of the sensing device and the diameter L**2** of the hole **301** may be changed to various numerical values depending on the target detection range of the sensing device.

[0070] For example, the diameter L**2** of the hole **301** of the cover **300** may be 9 mm, the distance L**1** between the hole **301** and the infrared sensor **100** may be 25 mm, the inner diameter L**3** at the rear end of the cover **300** or the diameter L**3** of the condenser lens **200** may be 26 mm. In this instance, the detection angle S of the infrared sensor **100** may be about 10.2°. The cover **300** may reduce the detection angle S of the infrared sensor **100** from about 120° down to about 10.2°. Additionally, the cover **300** may reduce the detection range of the infrared sensor **100** from Z**1**, Z**2**, Z**3** down to Z**1**. In this instance, D**1** may be about 10 m, and D**2** may be about 1.8 m or less.

[0071] According to the above-described configuration of the present disclosure, since the cover **300** limits the detection range of the infrared sensor **100** to the target detection range, it may be possible to increase accuracy of the infrared sensor **100**. Specifically, the sensing device does not detect a human entering Z**2** and Z**3** and only detects a human entering Z**10**, resulting in improved detection accuracy.

[0072] Referring to FIGS. **3** to **5**, the condenser lens **200** according to an embodiment of the present disclosure may have a protruding shape in the frontward direction or +X axis direction. For example, the condenser lens **200** may be a Fresnel lens having a convex shape in the frontward direction or +axis direction.

[0073] The cover **300** may have an internal space **302**. Additionally, the cover **300** may accommodate at least a portion of the condenser lens **200**. Additionally, the condenser lens **200** may be fixed, fit-assembled or fastened to the inside of the cover **300**.

[0074] According to the above-described configuration of the present disclosure, the cover **300** may be stably coupled to the condenser lens **200**. Additionally, the cover **300** may be replaceably coupled to the condenser lens **200**. Accordingly, the user may adjust the detection range of the sensing device by selectively coupling the cover **300** having various shapes to the condenser lens **200**.

[0075] Additionally, According to the above-described configuration of the present disclosure, the infrared sensor **100** may only detect infrared light passing through the hole **301**. The cover **300** may shield infrared light that does not pass through the hole **301** and enters the condenser lens **200** or the infrared sensor **100**. Accordingly, it may be possible to improve accuracy of the infrared sensor **100**.

[0076] FIG. **7** is a diagram showing the inner surface **303** of the cover **300** according to an embodiment of the present disclosure. Referring to FIG. **7**, the inner surface **303** of the cover **300** of the sensing device according to an embodiment of the present disclosure may have an emissive structure. The cover **300** may be configured to absorb infrared light directly incident on the inner surface **303** of the cover **300** through the hole **301**. Alternatively, the cover **300** may be configured not to reflect infrared light directly incident on the inner surface **303** of the cover **300** through the hole **301**. In this instance, the inner surface **303** of the cover **300** may have the emissivity of about 0.9 or more.

[0077] For example, the inner surface **303** of the cover **300** may be coated with a high emissivity material. The high emissivity material may include leather or black or matte paint.

[0078] Additionally, a film of high emissivity material may be attached to the inner surface **303** of the cover **300**.

[0079] Additionally, the inner surface **303** of the cover **300** may be processed to have high emissivity. For example, the emissivity of the inner surface **303** may be increased by sanding or chemical machining of the inner surface **303** of the cover **300**.

[0080] Additionally, the emissivity may be increased by forming recesses **310**, **320**, **330** or protrusions **310**, **320**, **330** on the inner surface **303** of the cover **300**. For example, the emissivity of the inner surface **303** may be increased by forming the recesses **310** in the shape of a rectangular hexahedron, the convex protrusions **320**, or the protrusions **330** in the shape of a triangular cone with a sharp tip on the inner surface **303** of the cover **300**.

[0081] According to the above-described configuration of the present disclosure, the cover **300** may only allow infrared light entering the condenser lens **200** as direct light through the hole **301** to pass through. The cover **300** may shield infrared light that is reflected from the inner surface **303** of the cover **300** and incident on the condenser lens **200**. Accordingly, it may be possible to increase detection accuracy of the infrared sensor **100**.

[0082] FIG. **8** is a partial perspective view of the sensing device according to an embodiment of the present disclosure. FIG. **9** is a cross-sectional perspective view of FIG. **8**, taken along the line B-B′. FIG. **10** is a cross-sectional view of FIG. **8**, taken along the line B-B′. Referring to FIGS. **8** to **10**, the cover **300** according to an embodiment of the present disclosure may include a shield wall **340** that protrudes from the inner surface **303** and covers the condenser lens **200**.

[0083] The shield wall **340** may be integrally formed with the cover **300**. Additionally, the shield wall **340** may be made of the same material as the cover **300**. Additionally, the shield wall **340** may be formed in the shape of a ring. The shield wall **340** may extend along the circumference of the hole **301** or the circumference of the condenser lens **200**.

[0084] The shield wall **340** may be located between the hole **301** and the condenser lens **200**. The shield wall **340** may be configured to cover at least a portion of the condenser lens **200**. Additionally, the shield wall **340** may shield at least some of incident infrared light through the hole **301**.

[0085] According to the above-described configuration of the present disclosure, the shield wall **340** may further limit the detection range of the infrared sensor **100**. Accordingly, it may be possible to further reduce the detection range of the infrared sensor **100**. The user may set the detection area using the sensing device of the present disclosure more precisely.

[0086] Additionally, according to the above-described configuration of the present disclosure, the shield wall **340** may shield infrared light that is reflected from the inner surface **303** of the cover **300** and incident on the condenser lens **200**. Accordingly, it may be possible to increase detection accuracy of the infrared sensor **100**.

[0087] Referring to FIGS. **8** to **10**, the shield wall **340** of the sensing device according to an embodiment of the present disclosure may include a shield hole **341** having a diameter L**5** that is smaller than the diameter L**2** of the hole **301**. At least some of infrared light entering the cover **300** through the hole **301** may be shielded by the shield wall **340**, and at least some may enter the condenser lens **200** through the shield hole **341**. The hole **301** and the shield hole **341** may have different sizes but the same shape. For example, the hole **301** and the shield hole **341** may have a

circular shape. Alternatively, the hole **301** and the shield hole **341** may have a polygonal shape.
[0088] According to the above-described configuration of the present disclosure, the shield wall **341** may further limit the detection range of the infrared sensor **100**. Accordingly, it may be possible to reduce the length of the cover **300** in the front-rear direction or the distance L**1** from the hole **301** to the infrared sensor **100**, and reduce the size of the sensing device.
[0089] FIG. **11** is a partial diagram of the sensing device according to an embodiment of the present disclosure. FIG. **12** is a diagram showing the detection range of the sensing device according to an embodiment of the present disclosure. Referring to FIGS. **11** and **12**, the cover **300** according to an embodiment of the present disclosure may be rotatable in the direction in which the infrared sensor **100** is oriented.
[0090] The direction in which the infrared sensor **100** is oriented may refer to a direction in which the sensing surface of the infrared sensor **100** is facing. Alternatively, the direction in which the infrared sensor **100** is oriented may refer to a direction perpendicular to the sensing surface of the infrared sensor **100**. Alternatively, the direction AX in which the infrared sensor **100** is oriented may refer to the frontward direction or +X axis direction. The cover **300** may be rotatable in the direction AX in which the infrared sensor **100** is oriented or an optical axis (AX direction) of the condenser lens **200**. That is, the direction AX in which the infrared sensor **100** is oriented or the AX direction of the condenser lens **200** may be a rotation axis AX of the cover **300**.
[0091] In this instance, a hole **301***a* of the cover **300** may have an asymmetrical shape. For example, the hole **301***a* may be approximately in the shape of a semicircle, an arc or a portion of a circle. Due to the shape of the hole **301***a*, when the hole **301***a* is circular in shape, the detection range of the infrared sensor **100** may be limited to a partial area. For example, when the hole **301***a* is circular in shape, the detection range may be Z**4** and Z**5**, but when the hole **301***a* is approximately semicircular in shape, the detection range may reduce to Z**4**.
[0092] In this instance, the user may adjust the detection range so that immobile objects such as walls, high voltage distribution boards, decorations, artworks and the like is included in the reduced detection range Z**5**.
[0093] Additionally, the detection range may be moved and adjusted by rotating the cover **300**. The rotation of the cover **300** may be manipulated by the user's manual work or a driving means such as a motor.
[0094] According to the above-described configuration of the present disclosure, the detection range of the infrared sensor **100** may be finely adjusted by rotating the cover **300**. Additionally, it may be possible to increase accuracy of the sensing device by excluding a detection needless range Z**5** from the detection range.
[0095] FIG. **13** is a diagram showing the sensing device according to an embodiment of the present disclosure. Referring to FIG. **13**, the sensing device according to an embodiment of the present disclosure may include the body **400**. At least one of the infrared sensor **100**, the condenser lens **200** or the cover **300** may be installed, fixed, coupled or fastened to the body **400**. The body **400** may form an appearance of the sensing device and provide an internal space. The body **400** may include an electronic component inside to operate the sensing device. For example, the body **400** may include a control unit and a power device inside. The power device may be connected to an external power source, and may have a battery or a solar module to supply power by itself. Additionally, the body **400** may include a plurality of parts.
[0096] The body **400** may have a warning unit **410**. The warning unit **410** may output sound or light when infrared light is detected through the infrared sensor **100**. For example, the warning unit **410** may provide a warning by outputting sound notifying that unauthorized access is prohibited or flashing light for warning.
[0097] Additionally, the warning unit **410** may be configured to select an alarm mode. For example, the alarm mode may include a day mode and a nigh mode. In the day mode, when infrared light is detected, the warning unit **410** may output sound. Additionally, in the night mode, when infrared

light is detected, the warning unit **410** may output flashing light without sound.

[0098] FIG. **14** is an exploded view of a configuration for installing the sensing device according to an embodiment of the present disclosure. FIG. **15** is a diagram showing an installation example of the sensing device according to an embodiment of the present disclosure. FIG. **16** is a diagram showing a plurality of sensing devices installed according to an embodiment of the present disclosure. Referring to FIGS. **13** to **16**, the sensing device according to an embodiment of the present disclosure may further include a coupling bar **430** extending below the body **400**. The coupling bar **430** may be formed on the lower surface of the body **400**, and extend in-Z axis direction or downward direction.

[0099] The coupling bar **430** may be coupled, inserted, fixed, fastened or assembled to the upper end or upper part of a coupling pipe **500**. The coupling pipe **500** may extend in the downward direction or-Z axis direction. The lower end or lower part of the coupling pipe **500** may be coupled, inserted, fixed, fastened, installed or assembled to a hole **801** of a traffic cone **800**. The coupling pipe **500** may include a rib **510** extending along the lengthwise direction of the coupling pipe **500**. The rib **510** may include a plurality of ribs **510**, and be arranged along the circumference of the coupling pipe **500**. The rib **510** may protrude in the circumferential direction of the coupling pipe **500** as it goes in +Z axis direction. Thereby, the outer diameter of the coupling pipe **500** may increase along +Z axis direction.

[0100] The traffic cone **800** may be a structure installed at construction sites, dangerous goods storages, roads and the like to indicate that the entry or passing of the corresponding areas are prohibited/restricted for a safety reason. The traffic cone **800** may be made of rubber or plastics, be hollow inside and have the hole **801** on top.

[0101] Additionally, a plurality of sensing devices may be installed at a regular interval. For example, the sensing device may be installed in each of a plurality of traffic cones **800**. The plurality of sensing devices makes it possible to set the wide detection range. In this instance, the plurality of sensing devices or the plurality of traffic cones **800** may be arranged at the interval D**3**. For example, D**3** may be set to about 1 to 7 m.

[0102] According to the above-described configuration of the present disclosure, the sensing device may be stably coupled, inserted, fixed, fastened, installed or assembled to the traffic cone **800**. Accordingly, it may be possible to install the sensing device by a relatively simple configuration.

[0103] FIG. **17** is a diagram showing the sensing device according to an embodiment of the present disclosure. Referring to FIG. **17**, the sensing device according to an embodiment of the present disclosure may further include a pair of sights **420** protruding from the upper surface of the body **400**.

[0104] The pair of sights **420** may be disposed along the X axis direction. An imaginary line connecting the pair of sights **420** may be parallel to the direction AX in which the infrared sensor **100** is oriented. The sights **420** may be configured to visually guide the detection direction AX of the infrared sensor **100** to the user.

[0105] According to the above-described configuration of the present disclosure, the user may easily install the sensing device using the sights **420**. When the sensing device is installed in a construction site or temporarily installed, the user may install the sensing device in a simple and accurate manner using the sights **420**. For example, when the sensing device is installed at the traffic cone **800**, the user may set the detection range by rotating the sensing device using the sights **420**.

[0106] FIG. **18** is an exploded view of a configuration for installing the sensing device according to an embodiment of the present disclosure. FIG. **19** is a diagram showing an installation example of the sensing device according to an embodiment of the present disclosure. FIG. **20** is an exploded view of a configuration for installing the sensing device according to an embodiment of the present disclosure. FIG. **21** is a diagram showing an installation example of the sensing device according to an embodiment of the present disclosure. Referring to FIGS. **18** to **21**, the sensing device according to an embodiment of the present disclosure may further include a mount **600**, **700** that is coupled to the coupling bar **430**.

[0107] One side of the mount **600**, **700** may be coupled, inserted, fixed, fastened or assembled to the coupling bar **430**. Additionally, the other side of the mount **600**, **700** may be coupled, inserted, fixed, fastened or assembled to a fixture such as a wall, a guardrail and an instrument. The mount **600**, **700** may be referred to as a bracket **600**, **700**, a wall mount **600**, **700** or a clamp **600**, **700**. The first mount **600** may be installed on the X-Y plane. Additionally, the second mount **700** may be installed on the Y-Z plane.

[0108] According to the above-described configuration of the present disclosure, the user may stably install the sensing device using the mount **600**, **700**.

[0109] Additionally, the sensing device according to an embodiment of the present disclosure may include a Velcro®. For example, the Velcro® may be disposed on one surface of the body **400**. Additionally, the Velcro® may be disposed on a structure or fixture at which the sensing device will be installed. The sensing device may be fixed or installed through the Velcro®.

[0110] Additionally, the sensing device according to an embodiment of the present disclosure may include a magnet. For example, the magnet may be disposed on one surface of the body **400**. Additionally, the magnet may be disposed on a structure or fixture at which the sensing device will be installed. The sensing device may be fixed or installed through the magnet.

[0111] The terms indicating directions such as upper, lower, left, right, front and rear are used for convenience of description, but it is obvious to those skilled in the art that the terms may change depending on the position of the stated element or an observer.

[0112] While the present disclosure has been hereinabove described with regard to a limited number of embodiments and drawings, the present disclosure is not limited thereto and it is apparent that a variety of changes and modifications may be made by those skilled in the art within the technical aspect of the present disclosure and the scope of the appended claims and their equivalents.

## Claims

**1**. A sensing device, comprising: an infrared sensor; a condenser lens located adjacent to a front side of the infrared sensor; and a cover located adjacent to a front side of the condenser lens and covering the condenser lens, the cover having a hole that is smaller than the condenser lens.

**2**. The sensing device according to claim 1, wherein the cover includes a portion having an inner diameter that increases from a front side thereof remote from the condenser lens toward a rear side thereof adjacent to the condenser lens.

**3**. The sensing device according to claim 1, wherein a distance between the hole of the cover and the infrared sensor is larger than a diameter of the hole.

**4**. The sensing device according to claim 1, wherein the cover extends rearwards from a front side thereof remove from the condenser lens to a rear side thereof adjacent to the condenser lens.

**5**. The sensing device according to claim 4, wherein an inner surface of the cover has an emissive structure recessed into or protruding from the inner surface.

**6**. The sensing device according to claim 4, wherein the condenser lens has a shape protruding frontward away from the infrared sensor, and the cover accommodates at least a portion of the condenser lens within an inner space thereof.

**7**. The sensing device according to claim 4, wherein the cover includes a shield wall protruding inward from an inner surface, the shield wall partially covering the condenser lens.

**8**. The sensing device according to claim 7, wherein the shield wall defines a shield hole therein having a smaller diameter than the hole of the cover.

**9**. The sensing device according to claim 1, wherein the cover is rotatable relative to the infrared sensor.

**10**. The sensing device according to claim 1, further comprising: a body into which the infrared sensor is installed; and a warning unit disposed in the body and configured to output sound or light in response to infrared light detected by the infrared sensor.
**11**. The sensing device according to claim 1, further comprising: a body into which the infrared sensor is installed; and a pair of sights protruding outwardly from an upper surface of the body.
**12**. The sensing device according to claim 1, further comprising: a body into which the infrared sensor is installed; and a coupling bar extending outwardly from a lower surface of the body.
**13**. The sensing device according to claim 12, further comprising: a mount that is coupled to the coupling bar.